

balloon_datalog

Pro Mini 3.3V / 8MHz



RED

system status

D9

heartbeat pulse
2 blinks + pause
3 blinks + pause
4 blinks + pause
5 blinks + pause

OK, main loop running
BMP180 fail
SD mount fail
SD open fail
SD write fail (runtime)



YELLOW

GPS status

D7

off
slow 1 Hz blink
fast 5 Hz blink
solid

no NMEA — module silent
getting bytes, no fix
fix but <6 sats
good fix ≥6 sats



GREEN

flight state

D5

off
2 Hz pulse
solid

WAITING (jumper in)
armed / flying
LANDED

field reference

self-test · pins · quick fixes

BOOT SEQUENCE

0 s	red LED heartbeat starts
~2 s	yellow starts 1 Hz (GPS bytes flowing)
30-180 s	yellow → solid (fix, ≥6 sats)
any	pull jumper → green pulses, log active

PIN MAP

D2	arm jumper ↔ GND	D7	yellow LED
D3	GPS TX (SoftSerial)	D9	red LED
D4	GPS RX (SoftSerial)	D10-13	SD (SPI, CS=10)
D5	green LED	A4/A5	BMP180 (I ² C)
D6	ESP32-CAM trigger	VCC/GND	3.3V power

QUICK FIXES

	3 red blinks SD card not mounted	→ reseal card, check FAT32/MBR
	4 red blinks SD open failed	→ delete FLIGHT.CSV, reformat
	yellow stays off GPS not sending bytes	→ check D3/D4 wiring + VCC
	yellow blinks 5+ min weak sky view	→ antenna ceramic-UP, open sky
	green won't pulse jumper not touching GND	→ verify D2 ↔ GND continuity